

How to make bubbles pop

The player will score points when the bubbles are popped, so the program has to make bubbles disappear from the screen. These next functions will allow it to do that.

8 This function will be used to remove a bubble from the game. It does this by deleting it from all the lists, and from the canvas. This code should be added directly after the code you typed out in step 7.

```
def del_bubble(i):
    del bub_r[i]
    del bub_speed[i]
    c.delete(bub_id[i])
    del bub_id[i]
```

This function deletes the bubble with ID "i"

Deletes the bubble from the radius and speed lists

Deletes the bubble from the canvas

Deletes the bubble from the ID list

9 Type this code to create a function that cleans up bubbles that have floated off the screen. This code should go directly after the code from step 8.

```
def clean_up_bubs():
    for i in range(len(bub_id)-1, -1, -1):
        x, y = get_coords(bub_id[i])
        if x < -GAP:
            del_bubble(i)
```

This goes through the bubble list backward to avoid the "for" loop causing an error when bubbles are deleted

Finds out where the bubble is

If the bubble is off the screen then it is deleted; otherwise, it would slow the game down

10 Now update the main game loop (from step 6) to include the helpful functions you have just created. Run it to make sure you haven't included any errors.

```
#MAIN GAME LOOP
while True:
    if randint(1, BUB_CHANCE) == 1:
        create_bubble()
    move_bubbles()
    clean_up_bubs()
    window.update()
    sleep(0.01)
```

Makes a new bubble

Updates the positions of all the bubbles

Redraws the window to show the changes

Removes bubbles that are off the screen



Don't forget to save your work

